# *Homocysteine, circulating vascular cell adhesion molecule and carotid atherosclerosis in postmenopausal vegetarian women and omnivores.*

Since the adoption of vegetarian diets as a healthy lifestyle has become popular, the cardiovascular effects of long-term vegetarianism need to be explored. The present study aimed to compare the presence and severity of carotid atherosclerosis (CA), and the blood levels of Vitamin B12, homocysteine (Hcy) and soluble vascular cell adhesion molecule-1 (sVCAM-1) between 57 healthy postmenopausal vegetarians and 61 age-matched omnivores. Carotid atherosclerosis, as measured by ultrasound, was found to be of no significant difference between the two groups. Yet, fasting blood glucose, low-density lipoprotein cholesterol, and Vitamin B12 were significantly lower, while Hcy and sVCAM-1 were higher in the vegetarians as comparing with the omnivores. Multivariate regression analysis showed that the level of Vitamin B12 was negatively associated with the level of Hcy. Vegetarianism itself and Hcy level were significantly associated with sVCAM-1 level in univariate analysis; however, after adjustment for covariates, we identified age but not vegetarianism as the determinant of sVCAM-1 level. Multiple linear regression analysis identified age and systolic blood pressure, but not vegetarianism, as determinants of common carotid artery IMT. In conclusion, there was no significant difference in CA between apparently healthy postmenopausal vegetarians and omnivores. The findings of elevated Hcy in vegetarians indicate the importance of prevention of Vitamin B12 deficiency.